

QUANTITATIVE INVESTMENT DECISIONS BASED ON MACHINE LEARNING AND INVESTOR ATTENTION ANALYSIS

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Abstract

According to the trading rules and financial data structure of the stock index futures market, and considering the impact of major emergencies, we intend to build a quantitative investment decision-making model based on machine learning. We first adopt the Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) signal decomposition technology to separate the short-term noise, cycle transformation and long-term trend from the original series, and use the CSI 500 Baidu index series to reflect the investors' attention, which provides data support for establishing a more effective forecasting model. Then, the CEEMDANBP neural network model is designed based on the obtained effective information of low-frequency trend series, investor attention index and CSI 500 stock index futures market transaction data. Finally, an Attention-based Dual Thrust quantitative trading strategy is

proposed and optimized. The optimized Attention-based Dual Thrust strategy solves the core problem of breakout interval determination, effectively avoids the risk of subjective selection, and can meet investors' different risk preferences. The quantitative investment decision-making model based on CEEMDAN-BP neural network utilizes the advantages of different algorithms, avoids some defects of a single algorithm, and can make corresponding adjustments according to changes in investors' attention and the occurrence of emergencies. The results show that considering investor attention can not only improve the predictive ability of the model, but also reduce the cognitive bias of the market, effectively control risks and obtain higher returns.

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Keywords: behavioral economics, decision making, signal decomposition, investor attention

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References

Anadu, K., Kruttli, M., McCabe, P., & Osambela, E. (2020). The shift from active to passive investing: Risks to financial stability? *Financial Analysts Journal*, 76(4), 23–39. <https://doi.org/10.1080/0015198X.2020.1779498>

Andrei, D., & Hasler, M. (2015). Investor attention and stock market volatility. *Review of Financial Studies*, 28(1), 33–72. <https://doi.org/10.1093/rfs/hhu059>

Audrino, F., Sigrist, F., & Ballinari, D. (2020). The impact of sentiment and attention measures on stock market volatility. *International Journal of Forecasting*, 36(2), 334–357. <https://doi.org/10.1016/j.ijforecast.2019.05.010>

Barber, B. M., & Odean, T. (2008). All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. *The Review of Financial Studies*, 21(2), 785–818. <https://doi.org/10.1093/rfs/hhm079>

Brooks, C., Rew, A. G., & Ritson, S. (2001). A trading strategy based on the lead-lag relationship between the spot index and futures contract for the FTSE 100. *International Journal of Forecasting*, 17(1), 31–44. [https://doi.org/10.1016/S0169-2070\(00\)00062-5](https://doi.org/10.1016/S0169-2070(00)00062-5)

Caldeira, J. F., & Moura, G. V. (2013). Selection of a portfolio of pairs based on cointegration: A statistical arbitrage strategy. *Brazilian Review of Finance*, 11(1), 49–80. <https://doi.org/10.12660/rbfin.v11n1.2013.4785>

Camerer, C. F., & Loewenstein, G. (2004). Behavioral economics: Past, present, future. In *Advances in behavioral economics* (pp. 3–51). Princeton University Press. <https://doi.org/10.1515/9781400829118-004>

Cao, J., Li, Z., & Li, J. (2019). Financial time series forecasting model based on CEEMDAN and LSTM. *Physica A: Statistical Mechanics and its Applications*, 519, 127–139. <https://doi.org/10.1016/j.physa.2018.11.061>

Chan, E. P. (2021). *Quantitative trading: How to build your own algorithmic trading business*. John Wiley & Sons.

Chen, C., Zhang, P., Liu, Y., & Liu, J. (2020). Financial quantitative investment using convolutional neural network and deep learning technology. *Neurocomputing*, 390, 384–390. <https://doi.org/10.1016/j.neucom.2019.09.092>

Covel, M. (2006). *Trend following: How great traders make millions in up or down markets*. FT Press.

Da, Z., Engelberg, J., & Gao, P. (2011). In search of attention. *The Journal of Finance*, 66(5), 1461–1499. <https://doi.org/10.1111/j.1540-6261.2011.01679.x>

Daniel, K., Hirshleifer, D., & Teoh, S. H. (2002). Investor psychology in capital markets: Evidence and policy implications. *Journal of Monetary Economics*, 49(1), 139–209. [https://doi.org/10.1016/S0304-3932\(01\)00091-5](https://doi.org/10.1016/S0304-3932(01)00091-5)

Deng, C., Zhou, X., Peng, C., & Zhu, H. (2022). Going green: Insight from asymmetric risk spillover between investor attention and pro-environmental investment. *Finance Research Letters*, 47, 102565. <https://doi.org/10.1016/j.frl.2021.102565>

De Prado, M. L. (2018). *Advances in financial machine learning*. John Wiley & Sons.

Fang, J., Gozgor, G., Lau, C. K. M., & Lu, Z. (2020). The impact of Baidu Index sentiment on the volatility of China's stock markets. *Finance Research Letters*, 32, 101099. <https://doi.org/10.1016/j.frl.2019.01.011>

García, F., Guijarro, F., Oliver, J., & Tamošiūnienė, R. (2018). Hybrid fuzzy neural network to predict price direction in the German DAX-30 index. *Technological and Economic Development of Economy*, 24(6), 2161–2178. <https://doi.org/10.3846/tede.2018.6394>

Guilbaud, F., & Pham, H. (2013). Optimal high-frequency trading with limit and market orders. *Quantitative Finance*, 13(1), 79–94. <https://doi.org/10.1080/14697688.2012.708779>

Han, L., Zhang, R., Wang, X., Bao, A., & Jing, H. (2019). Multi-step wind power forecast based on VMD-LSTM. *IET Renewable Power Generation*, 13(10), 1690–1700. <https://doi.org/10.1049/iet-rpg.2018.5781>

Henrique, B. M., Sobreiro, V. A., & Kimura, H. (2019). Literature review: Machine learning techniques applied to financial market prediction. *Expert Systems with Applications*, 124, 226–251. <https://doi.org/10.1016/j.eswa.2019.01.012>

Hirshleifer, D., & Teoh, S. H. (2003). Limited attention, information disclosure, and financial reporting. *Journal of Accounting and Economics*, 36(1–3), 337–386. <https://doi.org/10.1016/j.jacceco.2003.10.002>

- Huang, J., Li, Y., & Yao, H. (2022). Nonparametric mean-lower partial moment model and enhanced index investment. *Computers & Operations Research*, 144, 105814. <https://doi.org/10.1016/j.cor.2022.105814>
- Huang, N. E., Wu, Z., Long, S. R., Arnold, K. C., Chen, X., & Blank, K. (2009). On instantaneous frequency. *Advances in Adaptive Data Analysis*, 01(02), 177–229. <https://doi.org/10.1142/S1793536909000096>
- Hurst, B., Ooi, Y. H., & Pedersen, L. H. (2017). A century of evidence on trend-following investing. *The Journal of Portfolio Management*, 44(1), 15–29. <https://doi.org/10.3905/jpm.2017.44.1.015>
- Jiang, Y. (2022). Prediction model of the impact of innovation and entrepreneurship on China's digital economy based on neural network integration systems. *Neural Computing and Applications*, 34(4), 2661–2675. <https://doi.org/10.1007/s00521-021-05899-7>
- Ji, C., Zhang, C., Hua, L., Ma, H., Nazir, M. S., & Peng, T. (2022). A multi-scale evolutionary deep learning model based on CEEMDAN, improved whale optimization algorithm, regularized extreme learning machine and LSTM for AQI prediction. *Environmental Research*, 215, 114228. <https://doi.org/10.1016/j.envres.2022.114228>
- Kahneman, D. (1973). Attention and effort (Vol. 1063, pp. 218–226). Prentice-Hall.
- Kohzadi, N., Boyd, M. S., Kermanshahi, B., & Kaastra, I. (1996). A comparison of artificial neural network and time series models for forecasting commodity prices. *Neurocomputing*, 10(2), 169–181. [https://doi.org/10.1016/0925-2312\(95\)00020-8](https://doi.org/10.1016/0925-2312(95)00020-8)
- Kou, G., Chao, X., Peng, Y., Alsaadi, F. E., & Herrera-Viedma, E. (2019). Machine learning methods for systemic risk analysis in financial sectors. *Technological and Economic Development of Economy*, 25(5), 716–742. <https://doi.org/10.3846/tede.2019.8740>
- Krollner, B., Vanstone, B., & Finnie, G. (2010, April). Financial time series forecasting with machine learning techniques: A survey. In 18th European Symposium on Artificial Neural Networks (ESANN 2010): Computational Intelligence and Machine Learning (pp. 25–30). Bruges, Belgium.
- Kuang, Y., Singh, R., Singh, S., & Singh, S. P. (2017). A novel macroeconomic forecasting model based on revised multimedia assisted BP neural network model and ant Colony algorithm. *Multimedia Tools and Applications*, 76(18), 18749–18770. <https://doi.org/10.1007/s11042-016-4319-9>
- Li, X. (2006). Temporal structure of neuronal population oscillations with empirical model decomposition. *Physics Letters A*, 356(3), 237–241. <https://doi.org/10.1016/j.physleta.2006.03.045>
- Li, Y., Shen, D., Wang, P., & Zhang, W. (2020). Does intraday time-series momentum exist in Chinese stock index futures market? *Finance Research Letters*, 35, 101292. <https://doi.org/10.1016/j.frl.2019.09.007>
- Liu, F., Kang, Y., Guo, K., & Sun, X. (2021). The relationship between air pollution, investor attention and stock prices: Evidence from new energy and polluting sectors. *Energy Policy*, 156, 112430. <https://doi.org/10.1016/j.enpol.2021.112430>
- Lou, D. (2014). Attracting investor attention through advertising. *The Review of Financial Studies*, 27(6), 1797–1829. <https://doi.org/10.1093/rfs/hhu019>
- Makridakis, S., & Hibon, M. (1997). ARMA models and the Box-Jenkins methodology. *Journal of Forecasting*, 16(3), 147–163. [https://doi.org/10.1002/\(SICI\)1099-131X\(199705\)16:3<147::AID-FOR652>3.0.CO;2-X](https://doi.org/10.1002/(SICI)1099-131X(199705)16:3<147::AID-FOR652>3.0.CO;2-X)
- Mangram, M. E. (2013). A simplified perspective of the Markowitz Portfolio Theory. *Global Journal of Business Research*, 7(1), 59–70. <https://ssrn.com/abstract=2147880>

Markowitz, H. M. (1968). *Portfolio selection: Efficient diversification of investments*. Yale University Press.

Masini, R. P., Medeiros, M. C., & Mendes, E. F. (2021). Machine learning advances for time series forecasting. *Journal of Economic Surveys*, 1–36. <https://doi.org/10.1111/joes.12429>

Mbanga, C., Darrat, A. F., & Park, J. C. (2019). Investor sentiment and aggregate stock returns: The role of investor attention. *Review of Quantitative Finance and Accounting*, 53(2), 397–428. <https://doi.org/10.1007/s11156-018-0753-2>

Montoya-Cruz, E., Ramos-Requena, J. P., Trinidad-Segovia, J. E., & Sánchez-Granero, M. Á. (2020). Exploring arbitrage strategies in corporate social responsibility companies. *Sustainability*, 12(16), 1–17. <https://doi.org/10.3390/su12166293>

Mullainathan, S., & Thaler, R. H. (2000). *Behavioral economics* (NBER Working Paper No. 7948). <https://doi.org/10.3386/w7948>

Olgun, O., & Yetkiner, I. H. (2011). Determination of optimal hedging strategy for index futures: Evidence from Turkey. *Emerging Markets Finance and Trade*, 47(6), 68–79. <https://doi.org/10.2753/REE1540-496X470604>

Peng, L., & Xiong, W. (2006). Investor attention, overconfidence and category learning. *Journal of Financial Economics*, 80(3), 563–602. <https://doi.org/10.1016/j.jfineco.2005.05.003>

Pruitt, G., & Hill, J. R. (2012). *Building winning trading systems with Tradestation* (2nd ed.). (Wiley Trading Book 542). John Wiley & Sons. <https://doi.org/10.1002/9781119204954>

Rodriguez, D. (2020). *Backtrader*. <https://www.backtrader.com/>

Sampath, V. S., O'Connor, A. J., & Legister, C. (2022). Moral leadership and investor attention: An empirical assessment of the potus's tweets on firms' market returns. *Review of Quantitative Finance and Accounting*, 58(3), 881–910. <https://doi.org/10.1007/s11156-021-01012-0>

Sansa, N. A. (2020). The impact of the COVID-19 on the financial markets: Evidence from China and USA. *Electronic Research Journal of Social Sciences and Humanities*, 2(2), 29–39. <https://doi.org/10.2139/ssrn.3567901>

Schumaker, R. P., & Chen, H. (2009). A quantitative stock prediction system based on financial news. *Information Processing & Management*, 45(5), 571–583. <https://doi.org/10.1016/j.ipm.2009.05.001>

Sezer, O. B., Gudelek, M. U., & Ozbayoglu, A. M. (2020). Financial time series forecasting with deep learning: A systematic literature review: 2005–2019. *Applied Soft Computing*, 90, 106181. <https://doi.org/10.1016/j.asoc.2020.106181>

Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3), 425–442. <https://doi.org/10.1111/j.1540-6261.1964.tb02865.x>

Shen, D., Zhang, Y., Xiong, X., & Zhang, W. (2017). Baidu index and predictability of Chinese stock returns. *Financial Innovation*, 3(1), 4. <https://doi.org/10.1186/s40854-017-0053-1>

Siarni-Namini, S., & Namin, A. S. (2018). Forecasting economics and financial time series: ARIMA vs. LSTM. *arXiv*. <https://doi.org/10.48550/arXiv.1803.06386>

Smales, L. A. (2021). Investor attention and global market returns during the COVID-19 crisis. *International Review of Financial Analysis*, 73, 101616. <https://doi.org/10.1016/j.irfa.2020.101616>

Statcounter GlobalStats. (2020). Search Engine Market Share China. <https://gs.statcounter.com/search-engine-market-share/all/china/#yearly-2020-2020-bar>

Su, F., & Wang, X. (2021). Investor co-attention and stock return co-movement: Evidence from China's A-share stock market. *The North American Journal of Economics and Finance*, 58, 101548. <https://doi.org/10.1016/j.najef.2021.101548>

Sushko, V., & Turner, G. (2018). The implications of passive investing for securities markets. *BIS Quarterly Review*, 3, 113–131.

Szakmary, A. C., Shen, Q., & Sharma, S. C. (2010). Trend-following trading strategies in commodity futures: A re-examination. *Journal of Banking & Finance*, 34(2), 409–426. <https://doi.org/10.1016/j.jbankfin.2009.08.004>

Vozlyublennaia, N. (2014). Investor attention, index performance, and return predictability. *Journal of Banking & Finance*, 41, 17–35. <https://doi.org/10.1016/j.jbankfin.2013.12.010>

Yeh, J. R., Shieh, J. S., & Huang, N. E. (2010). Complementary ensemble empirical mode decomposition: A novel noise enhanced data analysis method. *Advances in Adaptive Data Analysis*, 2(02), 135–156. <https://doi.org/10.1142/S1793536910000422>

Yu, L., Wang, S., & Lai, K. K. (2008) Forecasting crude oil price with an EMD-based neural network ensemble learning paradigm. *Energy Economics*, 30(5), 2623–2635. <https://doi.org/10.1016/j.eneco.2008.05.003>

Zhang, D., Hu, M., & Ji, Q. (2020). Financial markets under the global pandemic of COVID-19. *Finance Research Letters*, 36, 101528. <https://doi.org/10.1016/j.frl.2020.101528>

Zhang, D., & Lou, S. (2021). The application research of neural network and BP algorithm in stock price pattern classification and prediction. *Future Generation Computer Systems*, 115, 872–879. <https://doi.org/10.1016/j.future.2020.10.009>

Zhang, Y., Chu, G., & Shen, D. (2021). The role of investor attention in predicting stock prices: The long short-term memory networks perspective. *Finance Research Letters*, 38, 101484. <https://doi.org/10.1016/j.frl.2020.101484>

Zhu, B., Wang, P., Chevallier, J., & Wei, Y. (2015). Carbon price analysis using empirical mode decomposition. *Computational Economics*, 45(2), 195–206. <https://doi.org/10.1007/s10614-013-9417-4>

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